Master’s Theorem

The Master Theorem applies to recurrences of the following form:

$T(n) = a\,T(n/b) + f(n)$ $\quad [f(n)=\Theta(n^k \log^p n)]$

$T(n) = a\,T(n-b) + f(n)$ $\quad [f(n)=O(n^k)]$

- $a \geq 1$, $b > 1$, $k>=0$ are constants and p is real number

For Decreasing Recursive Functions

$T(n) = a\,T(n-b) + f(n) \qquad [f(n)=O(n^k)]$

$a \geq 1$, $b \geq 1$, $k>=0$ are constants and p is real number

Defined on value of a

Case 1 : $a=1$

$T(n) = O(f(n)*n) = O(n^{k+1})$

Case 2: $a>1$

$T(n) = O(f(n)*a^{n/b}) = O(n^k\, a^{n/b})$

For Decreasing Recursive Functions

$$T(n) = a\,T(n-b) + f(n) \qquad [f(n)=O(n^k)]$$

$a \geq 1$, $b \geq 1$, $k>=0$ are constants and p is real number

Defined on value of a

Case 1 : a=1

$T(n) = O(f(n)*n) = O(n^{k+1})$

$T(n) = c + T(n-1)$ => $O(n)$

$= 1 + T(n-1)$ => $O(n)$

$= n + T(n-1)$ => $O(n^2)$

$= n^2 + T(n-1)$ => $O(n^3)$

$= \log n + T(n-1)$ => $O(n \log n)$

Case 2: a>1

$T(n) = O(f(n)*a^{n/b}) = O(n^k\, a^{n/b})$

$T(n) = 2T(n-1) + c$ => $O(2^n)$

$= 3T(n-1) + 1$ => $O(3^n)$

$= 2T(n-1) + n$ => $O(n2^n)$

The Master Theorem applies to recurrences of the following form:

$$T(n) = a\,T(n/b) + \Theta(n^k \log^p n)$$

$a \geq 1$, $b > 1$, $k >= 0$ are constants and p is real number

There are 3 cases: [compare $\log_b a$ and k]

1. If $a > b^k$

   then $T(n) = \Theta(n^{\log_b a})$.

2. If $a = b^k$

   a) If $p > -1$ then $T(n) = \Theta(n^{\log_b a} \log^{p+1} n)$

   b) If $p = -1$ then $T(n) = \Theta(n^{\log_b a} \log \log n)$

   c) If $p < -1$ then $T(n) = \Theta(n^{\log_b a})$

3. If $a < b^k$

   a) If $p >= 0$ then $T(n) = \Theta(n^k \log^p n)$

   b) a) If $p < 0$ then $T(n) = \Theta(n^k)$

$T(n) = a\,T(n/b) + \Theta(n^k \log^p n)$ [compare $\log_b a$ and k]

1.If $a > b^k$

then $T(n) = \Theta(n^{\log_b a})$.

2. If $a = b^k$

a)If $p > -1$ then $T(n) = \Theta(n^{\log_b a} \log^{p+1} n)$

b)If $p = -1$ then $T(n) = \Theta(n^{\log_b a} \log \log n)$

c)If $p < -1$ then $T(n) = \Theta(n^{\log_b a})$

3. If $a < b^k$

a)If $p >= 0$ then $T(n) = \Theta(n^k \log^p n)$

b)a) If $p < 0$ then $T(n) = \Theta(n^k)$

1)$T(n) = 4\,T(n/2) + n^2$ [a=4, b=2, k=2, p=0]

2)$T(n) = T(n/2) + n^2$ [a=1, b=2, k=2, p=0]

3)$T(n) = 2^n\,T(n/2) + n^n$ [Not applicable]

4)$T(n) = 14\,T(n/4) + n$ [a=14, b=4, k=1, p=0]

5)$T(n) = 2\,T(n/2) + n \log n$ [a=2, b=2, k=1, p=1]

6)$T(n) = 2\,T(n/2) + n / \log n$ [a=2, b=2, k=1, p=-1]

7)$T(n) = 2\,T(n/4) + n^{0.51}$ [a=2, b=4, k=0.51, p=0]

8)$T(n) = 0.5\,T(n/2) + 1/n$ [a=0.5, b=2, k=-1, p=0]

9)$T(n) = 6\,T(n/3) + n^2 \log n$ [a=6, b=3, k=2, p=1]

10)$T(n) = 64\,T(n/8) - n^2 \log n$ [a=64, b=8, k=2, p=1]

11)$T(n) = 7\,T(n/3) + n^2$ [a=7, b=3, k=2, p=0]

12)$T(n) = 4\,T(n/2) + \log n$ [a=4, b=2, k=0, p=1]

13)$T(n) = 2^{1/2}\,T(n/2) + n^2 \log n$ [a=$2^{1/2}$, b=2, k=2, p=1]

14)$T(n) = 2\,T(n/2) + n^{1/2}$ [a=2, b=2, k=1/2, p=0]

Video Lecture of the topic can be found at

Masters Theorem

https://youtu.be/d7WLeosewi8